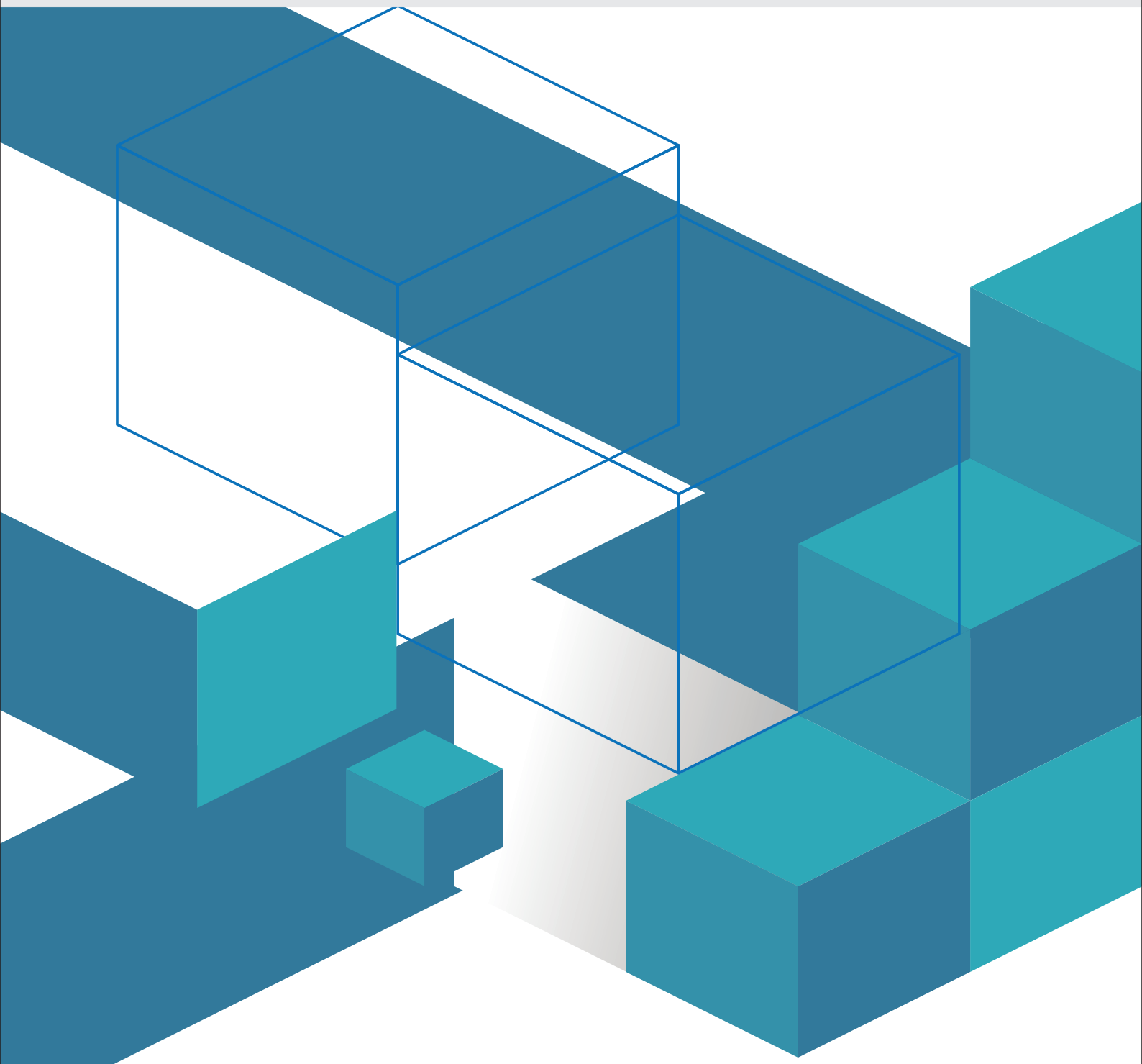
MATHEMATICS

DETERMINANT_MEGA_XII



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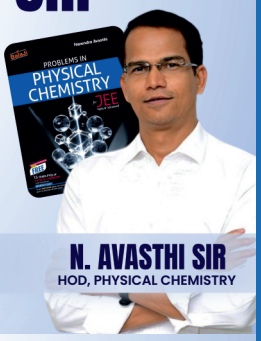

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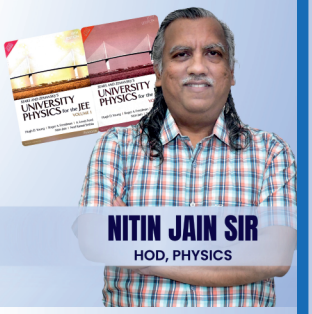


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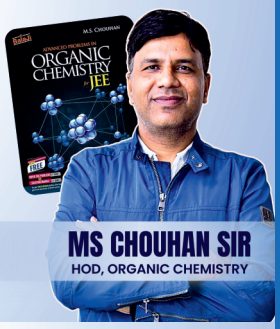


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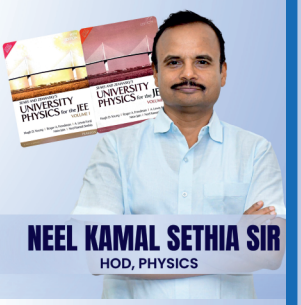


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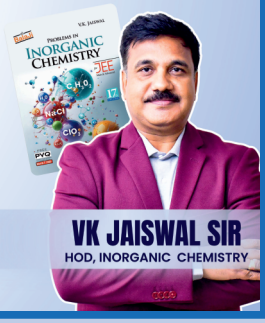


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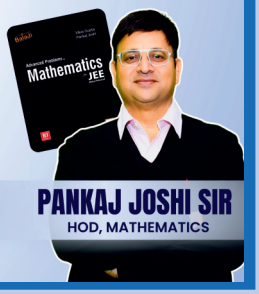


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MATHEMATICS

TARGET IIT JEE
MEGA-XII

DETERMINANT

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KEY CONCEPTS
DETERMINANT

1. The symbol $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is called the determinant of order two.

Its value is given by : $D = a_1 b_2 - a_2 b_1$

2. The symbol $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is called the determinant of order three .

Its value can be found as : $D = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$ **OR**

$D = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$ ..... and so on .

In this manner we can expand a determinant in 6 ways using elements of ;
R_1 , R_2 , R_3 or C_1 , C_2 , C_3 .

3. Following examples of short hand writing large expressions are :

(i) The lines : $a_1x + b_1y + c_1 = 0$..... (1)
$a_2x + b_2y + c_2 = 0$..... (2)
$a_3x + b_3y + c_3 = 0$..... (3)

are concurrent if , $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.

Condition for the consistency of three simultaneous linear equations in 2 variables.

(ii) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

(iii) Area of a triangle whose vertices are (x_r , y_r) ; r = 1 , 2 , 3 is :

$D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ If D = 0 then the three points are collinear .

(iv) Equation of a straight line passsing through (x_1 , y_1) & (x_2 , y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

4. MINORS :

The minor of a given element of a determinant is the determinant of the elements which remain after deleting the row & the column in which the given element stands . For example, the minor of a_1 in (Key Concept 2) is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order two will have “4 minors” & a determinant of order three will have “9 minors”.

5. COFACTOR :

If M_{ij} represents the minor of some typical element then the cofactor is defined as :
$C_{ij} = (-1)^{i+j} . M_{ij}$; Where i & j denotes the row & column in which the particular element lies. Note that the value of a determinant of order three in terms of ‘Minor’ & ‘Cofactor’ can be written as : $D = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$ **OR** $D = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ & so on .....

6. PROPERTIES OF DETERMINANTS :

P– 1 : The value of a determinant remains unaltered , if the rows & columns are inter changed.

e.g. if $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = D'$

D & D′ are transpose of each other . If $D' = -D$ then it is **SKEW SYMMETRIC** determinant but $D' = D \Rightarrow 2D = 0 \Rightarrow D = 0 \Rightarrow$ Skew symmetric determinant of third order has the value zero.

P– 2: If any two rows (or columns) of a determinant be interchanged , the value of determinant is changed in sign only. e.g.

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ & $D' = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ Then $D' = -D$.

P– 3: If a determinant has any two rows (or columns) identical , then its value is zero.

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ then it can be verified that D = 0.

P– 4: If all the elements of any row (or column) be multiplied by the same number, then the determinant is multiplied by that number.

e.g. If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ Then $D' = KD$

P–5: If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants . e.g.

$$\begin{vmatrix} a_1+x & b_1+y & c_1+z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

P– 6: The value of a determinant is not altered by adding to the elements of any row (or column) the same multiples of the corresponding elements of any other row

(or column). e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and

$D' = \begin{vmatrix} a_1+ma_2 & b_1+mb_2 & c_1+mc_2 \\ a_2 & b_2 & c_2 \\ a_3+na_2 & b_3+nb_2 & c_3+nc_2 \end{vmatrix}$. Then $D' = D$.

Note : that while applying this property **ATLEAST ONE ROW (OR COLUMN)** must remain unchanged .

P– 7: If by putting x = a the value of a determinant vanishes then $(x-a)$ is a factor of the determinant.

7. MULTIPLICATION OF TWO DETERMINANTS :

(i) $$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

(ii) If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$ then , $D^2 = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ where A_i , B_i , C_i are cofactors

PROOF : Consider $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = \begin{vmatrix} D & 0 & 0 \\ 0 & D & 0 \\ 0 & 0 & D \end{vmatrix}$

Note : $a_1A_2 + b_1B_2 + c_1C_2 = 0$ etc.

therefore , $D \times \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = D^3 \Rightarrow \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = D^2$ **OR** $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = D^2$

8. SYSTEM OF LINEAR EQUATION (IN TWO VARIABLES) :

(i) Consistent Equations : Definite & unique solution . [intersecting lines]

(ii) Inconsistent Equation : No solution . [Parallel line]

(iii) Dependent equation : Infinite solutions . [Identical lines]

Let $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ then :

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow$ Given equations are inconsistent

& $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ Given equations are dependent

9. CRAMER'S RULE : [Simultaneous Equations Involving Three Unknowns]

Let ,$a_1x + b_1y + c_1z = d_1$ ...(I) ; $a_2x + b_2y + c_2z = d_2$ ...(II) ; $a_3x + b_3y + c_3z = d_3$ ...(III)

Then , $x = \frac{D_1}{D}$, $Y = \frac{D_2}{D}$, $Z = \frac{D_3}{D}$.

Where $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$; $D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$; $D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ & $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

NOTE :

(a) If $D \neq 0$ and alteast one of $D_1, D_2, D_3 \neq 0$, then the given system of equations are consistent and have unique non trivial solution .

(b) If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$, then the given system of equations are consistent and have trivial solution only .

(c) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations are consistent and have infinite solutions.

In case $\left.\begin{array}{l} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{array}\right\}$ represents these parallel planes then also

$D = D_1 = D_2 = D_3 = 0$ but the system is inconsistent.

(d) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.

10. If x , y , z are not all zero , the condition for $a_1x + b_1y + c_1z = 0$; $a_2x + b_2y + c_2z = 0$ & $a_3x + b_3y + c_3z = 0$ to be consistent in x , y , z is that $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.

Remember that if a given system of linear equations have **Only Zero** Solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.

PROFICIENCY TEST-01

1. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ =

2. $\begin{vmatrix} 1 & 5 & \pi \\ \log_e e & 5 & \sqrt{5} \\ \log_{10} 10 & 5 & e \end{vmatrix}$ =

3. $\begin{vmatrix} 19 & 17 & 15 \\ 9 & 8 & 7 \\ 1 & 1 & 1 \end{vmatrix}$ =

4. The value of the determinant $\begin{vmatrix} 4 & -6 & 1 \\ -1 & -1 & 1 \\ -4 & 11 & -1 \end{vmatrix}$ is :

5. The value of the determinant $\begin{vmatrix} 31 & 37 & 92 \\ 31 & 58 & 71 \\ 31 & 105 & 24 \end{vmatrix}$ is :

6. $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$ =

7. $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ac \\ 1 & c & c^2-ab \end{vmatrix}$ =

8. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ =

9. The roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$ are

10. If $a \neq b \neq c$, the value of x (independent of a, b, c) which satisfies the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$, is:

11. If a + b + c = 0, then the solution of the equation $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$ is :

12. If $\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$, then x =

13. $\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix}$ =

14. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ =

15. $\begin{vmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{vmatrix}$ =

PROFICIENCY TEST-02

1. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} =$

2. $\begin{vmatrix} a+b & a+2b & a+3b \\ a+2b & a+3b & a+4b \\ a+4b & a+5b & a+6b \end{vmatrix} =$

3. $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} =$

4. The roots of the equation $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+x \end{vmatrix} = 0$ are

5. One of the roots of the given equation $\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$ is :

(A) $a + b + c$ (B) $-(a + b + c)$ (C) $a^2 + b^2 + c^2$ (D) $-(a^2 + b^2 + c^2)$

6. $\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix} =$

7. $\begin{vmatrix} 1/a & a^2 & bc \\ 1/b & b^2 & ca \\ 1/c & c^2 & ab \end{vmatrix} =$

8. $\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix} =$

9. $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix} =$

10. If $\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = k(x + y + z)(x - z)^2$, then k =

11. If –9 is a root of the equation $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ then the other two roots are :

12. If a, b, c are unequal what is the condition that the value of the following determinant is zero $\Delta = \begin{vmatrix} a & a^2 & a^3+1 \\ b & b^2 & b^3+1 \\ c & c^2 & c^3+1 \end{vmatrix}$

13. The value of the determinant $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$ is :

14. If a, b and c are non zero numbers, then $\Delta = \begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ is equal to :

15. If $\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -1 \end{vmatrix} = 0$, then the value of k is :

PROFICIENCY TEST-03

1. If $\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$, then $\begin{vmatrix} ka & kb & kc \\ kx & ky & kz \\ kp & kq & kr \end{vmatrix}$ =

2. If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$ = Ax – 12, then the value of A is :

3. The roots of the equation $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ are :

4. $\begin{vmatrix} \sin^2 x & \cos^2 x & 1 \\ \cos^2 x & \sin^2 x & 1 \\ -10 & 12 & 2 \end{vmatrix}$ =

5. If $D_P = \begin{vmatrix} p & 15 & 8 \\ p^2 & 35 & 9 \\ p^3 & 25 & 10 \end{vmatrix}$, then $D_1 + D_2 + D_3 + D_4 + D_5$ =

6. If $\begin{vmatrix} 4 & 1 \\ 2 & 1 \end{vmatrix}^2 = \begin{vmatrix} 3 & 2 \\ 1 & x \end{vmatrix} - \begin{vmatrix} x & 3 \\ -2 & 1 \end{vmatrix}$, then x =

7. If a, b, c are in A.P., then the value of $\begin{vmatrix} x+2 & x+3 & x+a \\ x+4 & x+5 & x+b \\ x+6 & x+7 & x+c \end{vmatrix}$ is :

8. If $\Delta = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix}$, then $\begin{vmatrix} x & 2y & z \\ 2p & 4q & 2r \\ a & 2b & c \end{vmatrix}$ equals

9. If $\begin{vmatrix} a & b & c \\ m & n & p \\ x & y & z \end{vmatrix} = k$, then $\begin{vmatrix} 6a & 2b & 2c \\ 3m & n & p \\ 3x & y & z \end{vmatrix}$ =

10. If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 5$; then the value of $\begin{vmatrix} b_2c_3 - b_3c_2 & c_2a_3 - c_3a_2 & a_2b_3 - a_3b_2 \\ b_3c_1 - b_1c_3 & c_3a_1 - c_1a_3 & a_3b_1 - a_1b_3 \\ b_1c_2 - b_2c_1 & c_1a_2 - c_2a_1 & a_1b_2 - a_2b_1 \end{vmatrix}$ is :

11. If $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = kabc(a+b+c)^3$, then the value of k is :

12. If A, B, C be the angles of a triangle, then $\begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$ =

13. $\begin{vmatrix} (a^x+a^{-x})^2 & (a^x-a^{-x})^2 & 1 \\ (b^x+b^{-x})^2 & (b^x-b^{-x})^2 & 1 \\ (c^x+c^{-x})^2 & (c^x-c^{-x})^2 & 1 \end{vmatrix} =$

14. The value of the determinant $\begin{vmatrix} 1 & \cos(\alpha-\beta) & \cos\alpha \\ \cos(\alpha-\beta) & 1 & \cos\beta \\ \cos\alpha & \cos\beta & 1 \end{vmatrix}$ is :

15. If $\begin{vmatrix} y+z & x-z & x-y \\ y-z & z+x & y-x \\ z-y & z-x & x+y \end{vmatrix}$ = kxyz, then the value of k is :

EXERCISE–I

1. (a) Prove that the value of the determinant $\begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$ is real.

(b) Prove that the value of the determinant $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} = 0$

(c) On which one of the parameter out of a, p, d or x, the value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ does not depend.

2. Without expanding as far as possible, prove that

(a) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$ (b) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$

3. If $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

4. Using properties of determinants or otherwise evaluate $\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$.

5. Prove that $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$.

6. If $D = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ and $D' = \begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix}$ then prove that $D' = 2D$.

7. Prove that $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$ = $(1 + a^2 + b^2)^3$.

8. Prove that $\begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix}$ = $(a + b + c)(a^2 + b^2 + c^2)$.

9. Show that the value of the determinant $\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$ vanishes for all values of A, B, C, P, Q & R where A + B + C + P + Q + R = 0

10. Factorise the determinant $\begin{vmatrix} bc & bc'+b'c & b'c' \\ ca & ca'+c'a & c'a' \\ ab & ab'+a'b & a'b' \end{vmatrix}$.

11. Prove that $\begin{vmatrix} (\beta+\gamma-\alpha-\delta)^4 & (\beta+\gamma-\alpha-\delta)^2 & 1 \\ (\gamma+\alpha-\beta-\delta)^4 & (\gamma+\alpha-\beta-\delta)^2 & 1 \\ (\alpha+\beta-\gamma-\delta)^4 & (\alpha+\beta-\gamma-\delta)^2 & 1 \end{vmatrix} = -64(\alpha-\beta)(\alpha-\gamma)(\alpha-\delta)(\beta-\gamma)(\beta-\delta)(\gamma-\delta)$

12. For a fixed positive integer n, if D= $\begin{vmatrix} n! & (n+1)! & (n+2)! \\ (n+1)! & (n+2)! & (n+3)! \\ (n+2)! & (n+3)! & (n+4)! \end{vmatrix}$ then show that $\left[\frac{D}{(n!)^3} - 4\right]$ is divisible by n.

13. Solve for x $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix}$ = 0.

14. If p + q + r = 0 , prove that $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$ = pqr $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$.

15. If a , b , c are all different & $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix}$ = 0, then prove that, abc(ab + bc + ca) = a + b + c.

16. Show that, $\begin{vmatrix} a^2+\lambda & ab & ac \\ ab & b^2+\lambda & bc \\ ac & bc & c^2+\lambda \end{vmatrix}$ is divisible by λ^2 and find the other factor.

17. (a) Without expanding prove that $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$.

(b) $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$.

18. Solve for x : $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix}$ = 0.

19. If $D_r = \begin{vmatrix} 2^{r-1} & 2(3^{r-1}) & 4(5^{r-1}) \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$ then prove that $\sum_{r=1}^{n} D_r = 0$.

20. In a Δ ABC, determine condition under which $\begin{vmatrix} \cot\frac{A}{2} & \cot\frac{B}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}+\tan\frac{C}{2} & \tan\frac{C}{2}+\tan\frac{A}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0$

21. Prove that : $\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$

22. Prove that $\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 \end{vmatrix} = 2(a_1-a_2)(a_2-a_3)(a_3-a_1)(b_1-b_2)(b_2-b_3)(b_3-b_1)$

23. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
and $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$,

then prove that $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{1/2}$ $(a, b, c \neq 0)$

24. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha-\beta)^2(\beta-\gamma)^2(\gamma-\alpha)^2$.

25. If $u = ax^2 + 2bxy + cy^2$, $u' = a'x^2 + 2b'xy + c'y^2$. Prove that

$$\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix} = \begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix} = -\frac{1}{y}\begin{vmatrix} u & u' \\ ax+by & a'x+b'y \end{vmatrix}.$$

EXERCISE–II

1. Solve using Cramer's rule : $\frac{4}{x+5} + \frac{3}{y+7} = -1$ & $\frac{6}{x+5} - \frac{6}{y+7} = -5$.

2. Solve the following using Cramer's rule and state whether consistent or not.

(a) $\begin{aligned} x+y+z-6&=0 \\ 2x+y-z-1&=0 \\ x+y-2z+3&=0 \end{aligned}$ (b) $\begin{aligned} 7x-7y+5z&=3 \\ 3x+y+5z&=7 \\ 2x+3y+5z&=5 \end{aligned}$

3. Solve the system of equations ; $\left.\begin{aligned} z+ay+a^2x+a^3&=0 \\ z+by+b^2x+b^3&=0 \\ z+cy+c^2x+c^3&=0 \end{aligned}\right]$

4. For what value of K do the following system of equations possess a non trivial (i.e. not all zero) solution over the set of rationals Q?
$x + Ky + 3z = 0$, $3x + Ky - 2z = 0$, $2x + 3y - 4z = 0$.
For that value of K , find all the solutions of the system.

5. Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$ where x , y , z are not all zero , prove that $a^2 + b^2 + c^2 + 2abc = 1$.

6. Given $a = \frac{x}{y-z}$; $b = \frac{y}{z-x}$; $c = \frac{z}{x-y}$ where x, y, z are not all zero, prove that: 1 + ab + bc + ca = 0.

7. If sin q ≠ cos q and x, y, z satisfy the equations

x cos p – y sin p + z = cos q + 1
x sin p + y cos p + z = 1 – sin q
x cos(p + q) – y sin (p + q) + z = 2

then find the value of $x^2 + y^2 + z^2$.

8. Investigate for what values of λ, μ the simultaneous equations x + y + z = 6; x + 2y + 3z = 10 & x + 2y + λz = μ have;
(a) A unique solution.
(b) An infinite number of solutions.
(c) No solution.

9. For what values of p, the equations : x + y + z = 1 ; x + 2y + 4z = p & x + 4y + 10z = p² have a solution ? Solve them completely in each case .

10. Solve the equations : Kx + 2y – 2z = 1, 4x + 2Ky – z = 2, 6x + 6y + Kz = 3 considering specially the case when K = 2.

11. Let a, b, c, d are distinct numbers to be chosen from the set {1, 2, 3, 4, 5}. If the least possible positive solution for x to the system of equations $\left.\begin{matrix} ax+by=1 \\ cx+dy=2 \end{matrix}\right\}$ can be expressed in the form $\frac{p}{q}$ where *p* and *q* are relatively prime, then find the value of (p + q).

12. If bc + qr = ca + rp = ab + pq = –1 show that $\begin{vmatrix} ap & a & p \\ bq & b & q \\ cr & c & r \end{vmatrix} = 0$.

13. If the following system of equations (a – t)x + by + cz = 0, bx + (c – t)y + az = 0 and cx + ay + (b – t)z = 0 has non–trivial solutions for different values of t, then show that we can express product of these values of t in the form of determinant .

14. Show that the system of equations

3x – y + 4z = 3 , x + 2y – 3z = –2 and 6x + 5y + λz = – 3

has atleast one solution for any real number λ. Find the set of solutions of λ = –5.

EXERCISE–III

1. If α, β, γ are the roots of $x^3 + px^2 + q = 0$, where q ≠ 0, then

$\Delta = \begin{bmatrix} 1/\alpha & 1/\beta & 1/\gamma \\ 1/\beta & 1/\gamma & 1/\alpha \\ 1/\gamma & 1/\alpha & 1/\beta \end{bmatrix}$ equals

(A) p/q (B) q/p (C) pq (D) 0

2. If $\Delta = \begin{vmatrix} \sqrt{6} & 2i & 3+\sqrt{6} \\ \sqrt{12} & \sqrt{3}+\sqrt{8}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{18} & \sqrt{2}+\sqrt{12}i & \sqrt{27}+2i \end{vmatrix}$ then Δ is ($i^2 = -1$)

(A) a negative integer (B) a natural number
(C) an irrational number (D) an imaginary number

3. If x, y, z are different from zero and

$\Delta = \begin{vmatrix} a & b-y & c-z \\ a-x & b & c-z \\ a-x & b-y & c \end{vmatrix} = 0$, then the value of the expression $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}$ is

(A) 0 (B) –1 (C) 1 (D) 2

4. If $p + q + r = a + b + c = 0$, then the determinant

$$\Delta = \begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$$ equals

(A) 0 (B) 1 (C) pa + qb + rc (D) abcpqr

5. If $p \neq a$, $q \neq b$, $r \neq c$ and the system of equations
$px + ay + az = 0$
$bx + qy + bz = 0$
$cx + cy + rz = 0$
has a non-trivial solution, then the value of

$$\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$$ is

(A) –1 (B) 0 (C) 1 (D) 2

6. If the system of linear equations $x + y + z = 6$, $x + 2y + 3z = 14$ and $2x + 5y + \lambda z = \mu$, ($\lambda, \mu \in R$) has more than one solution, then
(A) $\lambda \neq 8, \mu \in R$ (B) $\lambda = 8, \mu \neq 36$ (C) $\lambda = 8, \mu = 36$ (D) $\lambda = a, \mu \in R$

7. If a, b, c are positive and not all equal, then the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is always

(A) > 0 (B) 0 (C) < 0 (D) none of these

8. Let $px^4 + qx^3 + rx^2 + sx + t =$

$$\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$$ be an identity, where p, q, r, s, t are constants. Then t =

(A) 0 (B) 1 (C) 2 (D) –1

9. If $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \lambda \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, then $\lambda =$

(A) 1 (B) –1 (C) 2 (D) –2

10. The number of real values of λ for which the system of equations
$\lambda x + y + z = 0$, $x - \lambda y - z = 0$, $x + y - \lambda z = 0$ will have nontrivial solution is
(A) 0 (B) 1 (C) 2 (D) 3

11. If x, y, z be non-unity positive numbers, then

$$\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix} =$$

(A) 0 (B) 1 (C) –1 (D) value depends on x,y,z

12. $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$ if

(A) x, y, z are in A.P (B) x, y, z are in G.P (C) x, y, z are in H.P (D) xy, yz, zx are in A.P

13. The number of distinct real roots of the equation

$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is

(A) 0 (B) 1 (C) 2 (D) 3

14. The value of λ such that the system $x - 2y + z = -4$, $2x - y + 2z = 2$, $x + y + \lambda z = 4$ has infinitely many solutions is

(A) 0 (B) 1 (C) –1 (D) none of these

15. If the equations : $x + ay - z = 0$, $2x - y + az = 0$, $ax + y + 2z = 0$; have nontrivial solutions, then a can't be

(A) 2 (B) –2 (C) $1 + \sqrt{3}$ (D) $1 - \sqrt{3}$

EXERCISE–IV

1. If $a > 0$ and discriminant of $ax^2 + 2bx + c$ is –ve, then **[AIEEE 2002]**

$\begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ ax+b & bx+c & 0 \end{vmatrix}$ is equal to

(A) +ve (B) $(ac - b^2)(ax^2 + 2bx + c)$ (C) –ve (D) 0

2. If the system of linear equations **[AIEEE 2003]**

$x + 2ay + az = 0$; $x + 3by + bz = 0$; $x + 4cy + cz = 0$; has a non-zero solution, then a, b, c.

(A) Satisfy $a + 2b + 3c = 0$ (B) Are in A.P.

(C) Are in G.P. (D) Are in H.P.

3. If $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots\ldots$ are in G.P., then the value of the determinant **[AIEEE 2004]**

$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$, is

(A) –2 (B) 1 (C) 2 (D) 0

4. The system of equations **[AIEEE 2005]**

$\alpha x + y + z = \alpha - 1$

$x + \alpha y + z = \alpha - 1$

$x + y + \alpha z = \alpha - 1$

has infinite solutions, if α is

(A) –2 (B) Either –2 or 1 (C) not –2 (D) 1

5. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$, then f(x) is a polynomial of degree

(A) 1 (B) 0 (C) 3 (D) 2 **[AIEEE 2005]**

6. If $a_1, a_2, a_3, \ldots\ldots\ldots, a_n, \ldots\ldots\ldots$ are in G.P., then the determinant **[AIEEE 2005]**

$\Delta = \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ is equal to

(A) 1 (B) 0 (C) 4 (D) 2

7. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0$, $y \neq 0$, then D is **[AIEEE 2007]**

(A) Divisible by x but not y (B) Divisible by y but not x
(C) Divisible by neither x nor y (D) Divisible by both x and y

8. Let $A = \begin{vmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{vmatrix}$. If $|A^2| = 25$, then $|\alpha|$ equals. **[AIEEE 2007]**

(A) 1/5 (B) 5 (C) 5^2 (D) 1

9. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that x = cy + bz, y = az + cx, and z = bx + ay. Then $a^2 + b^2 + c^2 + 2abc$ is equal to **[AIEEE 2008]**

(A) 2 (B) –1 (C) 0 (D) 1

10. Let a, b, c be such that $b(a + c) \neq 0$ if **[AIEEE 2009]**

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0,$$ then the value of n is :

(A) Any even integer (B) Any odd integer (C) Any integer (D) Zero

11. Consider the system of linear equations : **[AIEEE 2010]**

$x_1 + 2x_2 + x_3 = 3$
$2x_1 + 3x_2 + x_3 = 3$
$3x_1 + 5x_2 + 2x_3 = 1$
Then system has

(A) Exactly 3 solutions (B) A unique solution (C) No solution (D) Infinite number of solutions

12. The number of values of k for which the linear equation $4x + ky + 2z = 0$, $kx + 4y + z = 0$ and $2x + 2y + z = 0$ posses a non-zero solution is **[AIEEE 2011]**

(A) 2 (B) 1 (C) Zero (D) 3

13. The number of values of k, for which the system of equations : **[JEE Main 2013]**

$(k + 1)x + 8y = 4k$
$kx + (k + 3)y = 3k - 1$
has no solution, is :

(A) 3 (B) infinite (C) 1 (D) 2

14. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then K is equal to :

(A) -1 (B) $\alpha\beta$ (C) $\frac{1}{\alpha\beta}$ (D) 1 **[JEE Main 2014]**

15. The set of all values of λ for which the system of linear equations : **[JEE Main 2015]**

$2x_1 - 2x_2 + x_3 = \lambda x_1$

$2x_1 - 3x_2 + 2x_3 = \lambda x_2$

$-x_1 + 2x_2 = \lambda x_3$

has a non-trivial solution,

(A) contains more than two elements (B) is an empty set

(C) is a singleton (D) contains two elements

16. The system of linear equations **[JEE Main 2016]**

$x + \lambda y - z = 0$

$\lambda x - y - z = 0$

$x + y - \lambda z = 0$

has a non-trivial solution for :

(A) infinitely many values of λ. (B) exactly one value of λ

(C) exactly two values of λ (D) exactly three values of λ

17. If S is the set of distinct values of 'b' for which the following system of linear equations

$x + y + z = 1$

$x + ay + z = 1$

$ax + by + z = 0$

has no solution, then S is : **[JEE Main 2017]**

(A) a finite set containing two or more elements (B) a singleton

(C) an empty set (D) an infinite set

18. If the system of linear equations

$x + ky + 3z = 0$

$3x + ky - 2z = 0$

$2x + 4y - 3z = 0$

has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to **[JEE Main 2018]**

(A) 30 (B) -10 (C) 10 (D) -30

19. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then the ordered pair (A, B) is equal to **[JEE Main 2018]**

(A) (4, 5) (B) (– 4, – 5) (C) (– 4, 3) (D) (– 4, 5)

20. Let S be the set of all real values of k for which the system of linear equations

$x + y + z = 2$

$2x + y - z = 3$

$3x + 2y + kz = 4$

has a unique solution. Then S is : **[JEE Main 2018]**

(A) an empty set (B) equal to {0} (C) equal to R (D) equal to R–{0}

21. If the system of linear equations

$x + ay + z = 3$

$x + 2y + 2z = 6$

$x + 5y + 3z = b$

has no solution, then : **[JEE Main 2018]**

(A) $a = -1, b = 9$ (B) $a = -1, b \neq 9$ (C) $a \neq -1, b = 9$ (D) $a = 1, b \neq 9$

EXERCISE–V

1. (a) If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$ then $f(100)$ is equal to :

(A) 0 (B) 1 (C) 100 (D) -100

(b) Let a, b, c, d be real numbers in G.P. If u, v, w satisfy the system of equations,

$u + 2v + 3w = 6$

$4u + 5v + 6w = 12$

$6u + 9v = 4$

then show that the roots of the equation,

$$\left(\frac{1}{u} + \frac{1}{v} + \frac{1}{w}\right) x^2 + [(b-c)^2 + (c-a)^2 + (d-b)^2]x + u + v + w = 0$$

and $20x^2 + 10(a-d)^2 x - 9 = 0$

are reciprocals of each other. **[JEE '99, 2+10 out of 200]**

2. If the system of equations $x - Ky - z = 0$, $Kx - y - z = 0$ and $x + y - z = 0$ has a non zero solution, then the possible values of K are **[JEE 2000 (Screening)]**

(A) $-1, 2$ (B) $1, 2$ (C) $0, 1$ (D) $-1, 1$

3. Prove that for all values of θ, $\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} = 0$

[JEE 2000 (Mains), 3 out of 100]

4. Find the real values of r for which the following system of linear equations has a non-trivial solution. Also find the non-trivial solutions :

$2rx - 2y + 3z = 0$

$x + ry + 2z = 0$

$2x + rz = 0$ **[REE 2000 (Mains) , 3 out of 100]**

5. Solve for x the equation

$$\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0$$ **[REE 2001 (Mains) , 3 out of 100]**

6. Test the consistency and solve them when consistent, the following system of equations for all values of λ

$x + y + z = 1$

$x + 3y - 2z = \lambda$

$3x + (\lambda + 2)y - 3z = 2\lambda + 1$ **[REE 2001 (Mains) , 5 out of 100]**

7. Let a, b, c be real numbers with $a^2 + b^2 + c^2 = 1$. Show that the equation

$$\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$$

represents a straight line. **[JEE 2001 (Mains) , 6 out of 100]**

8. The number of values of k for which the system of equations

$$(k + 1)x + 8y = 4k$$
$$kx + (k + 3)y = 3k - 1$$

has infinitely many solutions is **[JEE 2002 (Screening), 3]**

(A) 0 (B) 1 (C) 2 (D) inifinite

9. The value of λ for which the system of equations $2x - y - z = 12$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ has no solution is **[JEE 2004 (Scr.)]**

(A) 3 (B) –3 (C) 2 (D) –2

10. (a) Consider three points P = $\left(-\sin(\beta-\alpha), -\cos\beta\right)$, Q = $\left(\cos(\beta-\alpha), \sin\beta\right)$ and R = $\left(\cos(\beta-\alpha+\theta), \sin(\beta-\theta)\right)$, where $0 < \alpha, \beta, \theta < \pi/4$

(A) P lies on the line segment RQ (B) Q lies on the line segment PR

(C) R lies on the line segment QP (D) P, Q, R are non collinear

(b) Consider the system of equations

$$x - 2y + 3z = -1$$
$$-x + y - 2z = k$$
$$x - 3y + 4z = 1$$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True **[JEE 2008, 3 + 3]**

11. Which of the following values of α satisfy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$?

[IIT Advance - 2015]

(A) –4 (B) 9 (C) –9 (D) 4

12. The total number of distinct $x \in \mathbf{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is : **[IIT Advance - 2016]**

13. Let $\alpha, \lambda, \mu \in \mathbf{R}$. Consider the system of linear equations **[IIT Advance - 2016]**

$$\alpha x + 2y = \lambda$$
$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct?

(A) If $\alpha = -3$, then the system has infinitely many solutions for all values of λ and μ.

(B) If $\alpha \neq -3$, then the system has a unique solution for all values of λ and μ.

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $\alpha = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $\alpha = -3$

14. Let p, q, r be nonzero real numbers that are, respectively, the 10th, 100th and 1000th terms of a harmonic progression. Consider the system of linear equations

$$x + y + z = 1$$
$$10x + 100y + 1000z = 0$$
$$qr\,x + pr\,y + pq\,z = 0$$

List-I	**List-II**
(I) If $\frac{q}{r} = 10$, then the system of linear equations has	(P) $x = 0,\ y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II) If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10$, then the system of linear equations has	(S) no solution
	(T) at least one solution

The correct option is : **[JEE Advanced 2022]**

(A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)
(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)
(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)
(D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

15. Let α, β and γ be real numbers. Consider the following system of linear equations

$x + 2y + z = 7$
$x + \alpha z = 11$
$2x - 3y + \beta z = \gamma$

Match each entry in List-I to the correct entries in List-II.

List-I	**List-II**
(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has	(1) a unique solution
(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $y \neq 28$, then the system has	(2) no solution
(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $y \neq 28$, then the system has	(3) infinitely many solutions
(S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has	(4) $x = 11,\ y = -2$ and $z = 0$ as a solution
	(5) $x = -15,\ y = 4$ and $z = 0$ as a solution

The correct option is : **[JEE Advanced 2023]**

(A) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (1) (S) $\rightarrow$ (4)
(B) (P) $\rightarrow$ (3) (Q) $\rightarrow$ (2) (R) $\rightarrow$ (5) (S) $\rightarrow$ (4)
(C) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (4) (S) $\rightarrow$ (5)
(D) (P) $\rightarrow$ (2) (Q) $\rightarrow$ (1) (R) $\rightarrow$ (1) (S) $\rightarrow$ (3)

ANSWER KEY

PROFICIENCY TEST-01

1. (a – b) (b – c) (c – a) **2.** 0 **3.** 0 **4.** –25 **5.** 0 **6.** 0

7. 0 **8.** xy **9.** x = 2, –1 **10.** x = 0 , x = $\pm\sqrt{\sum ab}$

11. x = 0, $\pm\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$

12. 1,–9 **13.** $1 + a^2 + b^2 + c^2$ **14.** (a – b) (b – c) (c – a) (a + b + c) **15.** 0

PROFICIENCY TEST-02

1. $-(a^3 + b^3 + c^3 - 3abc)$ **2.** 0 **3.** 4abc **4.** 0, –3 **5.** B **6.** –2

7. 0 **8.** $4a^2b^2c^2$ **9.** $xyz\left(1+\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ **10.** 1 **11.** 2, 7

12. abc = –1 **13.** 0 **14.** 0 **15.** $\frac{33}{8}$

PROFICIENCY TEST-03

1. $k^3\Delta$ **2.** 24 **3.** {–9, 0, 9} **4.** 0 **5.** –28000 **6.** 6 **7.** 0
8. 4Δ **9.** 6k **10.** 25 **11.** 2 **12.** 0 **13.** 0 **14.** 0
15. 8

EXERCISE–I

1. (c) p **4.** -1 **10.** $(ab' - a'b)(bc' - b'c)(ca' - c'a)$ **13.** $x = -1$ or $x = -2$

16. $\lambda^2 (a^2 + b^2 + c^2 + \lambda)$ **18.** x = 4 **20.** Triangle ABC is isosceles.

EXERCISE–II

1. $x = -7 , y = -4$ **2.** (a) x = 1 , y = 2 , z = 3 ; consistent (b) inconsistent

3. x = –(a + b + c) , y = ab + bc + ca , z = –abc **4.** K = $\frac{33}{2}$, $x : y : z = -\frac{15}{2} : 1 : -3$ **7.** 2

8. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$

9. $x = 1 + 2K , y = -3K , z = K$, when p = 1 ; x = 2K, $y = 1 - 3K , z = K$ when p = 2 ; where $K \in R$

10. If $K \neq 2, \frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2\left(K^2+2K+15\right)}$, If K= 2, then $x = \lambda, y = \frac{1-2\lambda}{2}$ and z = 0 where $\lambda \in R$

11. 19 **13.** $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

14. If $\lambda \neq -5$ then $x = \frac{4}{7}$; $y = -\frac{9}{7}$ and z = 0 ;

If $\lambda = 5$ then $x = \frac{4-5K}{7}$; $y = \frac{13K-9}{7}$ and $z = K$ where $K \in R$

EXERCISE–III

1.	D	**2.**	A	**3.**	D	**4.**	A	**5.**	D	**6.**	C	**7.**	C
8.	A	**9.**	B	**10.**	B	**11.**	A	**12.**	B	**13.**	B	**14.**	D
15.	A												

EXERCISE–IV

1.	C	**2.**	D	**3.**	D	**4.**	D	**5.**	D	**6.**	B	**7.**	D
8.	A	**9.**	D	**10.**	B	**11.**	C	**12.**	A	**13.**	C	**14.**	D
15.	D	**16.**	D	**17.**	B	**18.**	C	**19.**	D	**20.**	D	**21.**	B

EXERCISE–V

1. **(a)** A **2.** D **4.** $r = 2$; $x = k$; $y = k/2$; $z = -k$ where $k \in R - \{0\}$

5. $x = n\pi$, $n \in I$

6. If $\lambda = 5$, system is consistent with infinite solution given by $z = K$,

$y = \frac{1}{2}(3K + 4)$ and $x = -\frac{1}{2}(5K + 2)$ where $K \in R$

If $\lambda \neq 5$, system is consistent with unique solution given by $z = \frac{1}{3}(1 - \lambda)$; $x = \frac{1}{3}(\lambda + 2)$ and $y = 0$.

8.	B	**9.**	D	**10.**	(a) D; (b) A	**11.**	BC	**12.**	2	**13.**	BCD
14.	B	**15.**	A								